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1.0 INTRODUCTION

Safety & Health issues has got an attention in the developed countries since the beginning of industrialization or the advent of mechanization into work processes, it is relatively a recent practice in most developing countries among which our country, Ethiopia is the one. This is the fact that industrialization is a late development endeavor in this part of the world.

Today experience has clearly shown that the growth of each and every company inevitably depends on the quality of working conditions and the wellbeing of working population. That means the health & safety of the working population has a bilateral relationship with the productivity of the company. Since workers become healthy physically, mentally & socially they spend most of their time at work, absenteeism and other medical costs will be decreased. Thus in order to ascertain company goals and objectives, it is necessary to see them in relation to maintenance of safe working conditions.

Currently Safety & Health Issues has also become one of the four main components of the construction industry i.e. cost, quality, time and safety & health. So, to become productive and competitive in the industry it is crucial to have sound safety & health policy and enter into action. Aser construction PLC as one of Grade-1 general contractor develops this policy for the realization of good safety culture in the company. Therefore, all concerned bodies and employees throughout the company should give emphasis, follow and practice this policy for the development of good safety culture.

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2.0 Purpose

The introduction of this Safety and Health working rule aims to improve the current working environment in general. Specifically, the procedures seek, among others, to:

- 2.1 Create safe working environment and minimize occupational hazards and accidents,
- 2.2 Prevent employees from environmental and work related health hazards,
- 2.3 Create safe system of work and guide staff and management in the administration and control of occupational hazards,
- 2.4 Establish safety culture (i.e. to create *safe person at work*) throughout the company,
- 2.5 Train and guide staff in undertaking their responsibilities safely,
- 2.6 Protect the health, safety and welfare of employees and other people who may be affected by the work of the organization - with special attention to female employees,
- 2.7 Ensure uniformity, consistency and continuity of safety and health management, principles and practice,
- 2.8 Reduce and control the cost & pain of accidents to the employees and the firm,
- 2.9 Coordinate the activities of the safety and health function with other functions, transport, machinery, human resources and construction in particular.
- 2.10 This Safety and Health working rule shall apply to all Head Office and Project Sites as well as all sub-contractors and consultants.
- 2.11 The Manual should be read in conjunction with other Policy procedure manuals and Working rules; Construction Equipment's, Engineering and Human Resources in particular.
- 2.12 In case of clarity about any particular clause, sentence or word contained in this Manual, the interpretation of the General Manager/ Deputy General Manager of the company shall be final.
- 2.13 The General Manager/ Deputy General Manager is vested with the authority to introduce additional working rules as well as to amend or alter existing Procedures, Working rules etc. as may be required from time to time. He shall consult with his management staff when necessary.

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3.0 The need for formulating Safety and Health Systems

3.1 The main objectives of formulating Safety and Health Systems are:

- 3.1.1 The alarmingly increasing accidents at project sites, machinery and equipment operations signal the need to establish standard in the area;
- 3.1.2 Health and Safety issues are the prime management responsibilities of the Company ranking equally with responsibilities for construction, costs, Quality and similar matters;
- 3.1.3 The company is committed to operating the business in accordance with modern safety and health rules and all applicable regulations made by the government and concerned bodies as reasonably practicable;
- 3.1.4 To ensure that it is the duty of management to see that everything reasonably practicable is done to prevent personal injury in the processes of all operations of all site, machinery and equipment, and to maintain a safe and healthy place of work;
- 3.1.5 To inform employees that it is their duty to act responsibly, and to do everything they can to prevent injury to themselves and fellow workers.

4.0 SAFETY AND HEALTH FUNCTION

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4.1 Safety and Health Management Organization

4.1.1 The Safety and Health Function in the company shall be organized under Human resource development& Administration Department (HRDAD) and handled by Safety & Health officer. The Safety & Health officer will have functional responsibility to Construction Department with regard to Safety and Health.

4.1.2 At project sites the Safety and Health affairs shall be handled by the Head, Project Administration Section (HPAS).

4.1.3 Occupational Safety and Health Committee (OSHC) shall be established at the Head Office, It consist a minimum of five members

- a. HRD&AdministrationDepartment Manager Chairperson
- b. Construction Department Manager Member
- c. Equipment & Vehicles MaintenanceDepartment Member
Manager
- d. Finance Department Manager Member
- e. Procurement and Supply Department Manager Member
- f. Safety & Health Officer Secretary

4.1.4 Site Occupational Safety and Health Committee (SOSHC) will be established at project level consisting a minimum of five members.

- a. Head, ProjectAdministration Section Chairperson
- b. Head, Project Construction Section Member
- c. Head, ProjectEquipmentMaintenance Member
Section
- d. Head, ProjectSupply Section Member

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e. Site Engineer /General Forman

Member

f. Safety and Health Officer

Secretary

4.2 ASSIGNMENTS of Responsibility

4.2.1 Safety & Health Officer (SHO) at HO

The Head Office Safety & Health Officer (SHO) is responsible for:

4.2.1.1 Coordinating the overall safety and health program of the company.

4.2.1.2 Ensuring that the Health and Safety working rule of the company is implemented.

4.2.1.3 Identifying the applicable standards, guidelines, and recommendations necessary for a safe and healthy work place.

4.2.1.4 Assisting the OSHC in the development and updating of standard operating procedures.

4.2.1.5 Reviewing the Health and Safety working rule Manual as necessary.

4.2.1.6 Comprehensively inspecting all the company work places against appropriate standards.

4.2.1.7 Identifying the need for, and providing, training and orientation.

4.2.1.8 Identifying the need for health and safety resources jointly with the concerned departments.

4.2.1.9 Providing consultation on safety and health emergencies.

4.2.1.10 Ensuring comprehensive records of problems and solutions are maintained at all levels and shared this information with the appropriate supervisor and safety and health committee.

4.2.1.11. Preparing monthly and annual reports.

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4.2.2 The Head, Project Administration Section (HPAS)

The Head, Project Administration Section (HPAS) is responsible for: -

4.2.2.1. Coordinating the site safety and health program.

4.2.2.2. Ensuring that the Health and Safety working rule of the company is implemented.

4.2.2.3. Assisting in the identification of applicable standards, guidelines, and recommendations necessary for a safe and healthy work place.

4.2.2.4. Maintaining comprehensive records of problems and solutions at site level and providing this information to the appropriate supervisors, SOSHC and OSH Committee at HO.

4.2.2.5 Preparing monthly and annual reports

4.2.3 Managers and Supervisors

Managers and Supervisors at all levels throughout the organization are responsible for maintaining a safe and healthy workplace by:

4.2.3.1 Complying with all applicable safety and health standards, rules, regulations and orders issued by the company and the Labor Proclamation pertaining to the activities immediately under their jurisdiction;

4.2.3.2. Ensuring that employees are instructed and/or trained in safe practices and methods of job performance as pertains to their assignment;

4.2.3.3. Ensuring that sick and/or injured employees performing official duties receive appropriate first aid and/or further treatment as necessarily;

4.2.3.4. Investigating and reporting each incident and/or injury in accordance with established procedures;

4.2.3.5. Initiating, to the limit of their authority and capability, such actions necessary to correct unsafe or unhealthful working conditions determined to exist and promptly advising appropriate management when such conditions require corrective actions beyond their authority;

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4.2.3.6 Conducting regular surveys of their operations to ensure compliance with such safety standards, codes, regulations, rules and orders applicable to the work area concerned;

4.2.3.7 Ensuring that employees under their supervision are aware of their responsibilities;

3.2.3.8 Taking immediate action to correct any violation of safety rules observed or reported to them.

4.2.3.9. Planning the workload and assigning employees to jobs, which they are qualified to perform.

4.2.3.10. Ensuring that the employees understand the work to be done, the hazards that may be encountered, and the proper working rule for doing the work safely;

4.2.3.11. Ensuring that required guards and protective equipment are provided, used, and properly maintained;

3.2.3.12. Ensuring that all tools, equipment, Machineries and Vehicles are properly maintained, inspected and used;

4.2.3.13. Ensuring safe working conditions.

4.2.4 Employees

Employees at all levels throughout the Company are responsible for:

4.2.4.1. Complying with such occupational health and safety standards, rules, regulations, and orders issued by the company and the provisions of the Labor Law are applicable to an employee's action and conduct;

4.2.4.2. Promptly advising their supervisor regarding all work related incidents resulting in personal injury, illness and/or property damage;

4.2.4.3 Promptly report to their supervisor or Safety and Health Officer / Assistant any unsafe or unhealthful conditions in the work environment;

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4.2.4.4 Taking all necessary and appropriate safety precautions to protect themselves, other personnel and the environment;

4.2.4.5 Using properly all safeguards, safety devices and other appliances furnished for the protection of health and safety of themselves and others;

4.2.4.6 obeying all safety and health instructions issued by the concerned personnel;

4.2.4.7 Taking all necessary and appropriate safety precautions to protect themselves from exposure to HIV/AIDS and addiction to alcohol and drugs including “Chat”.

4.2.5 The Head Office Occupational Safety and Health Committee (OSHC)

The Occupational Safety and Health Committee (OSHC) at the Head Office shall be responsible to:

4.2.5.1 Provide recommendations to the General Manager and/or the Management Committee regarding health and safety issues;

4.2.5.2 Identify unsafe work practices and conditions and suggest appropriate remedies;

4.2.5.3 Ensure that employee and others (Visitors, Consultants, Subcontractors, etc., as applicable) are informed about safety policies, training programs, injury risks and causation, and other health and safety-related matters;

4.2.5.4 Maintain an open channel of communication between employees and management concerning occupational and environmental health and safety matters;

4.2.5.5 Provide a means by which employees can utilize their knowledge of workplace operations to advise management in the improvement of policies, condition, and practices;

4.2.5.6 Review existing health and safety policies;

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4.2.5.7 Review health and safety resource requirements and recommend priorities for allocation;

4.2.5.8 Undertake Safety Survey and Audit to monitor quality and progress of the Health and Safety System;

4.2.5.9 Review monthly occupational safety and health report;

4.2.5.10 Hold meetings scheduled by the Manager of Administration and HRD Department or the assigned chairperson of the OSHC at least monthly.

4.2.6 The Site Occupational Safety and Health Committee (SOSHC)

The Site Occupational Safety and Health Committee (SOSHC) at each project is responsible to:

4.2.6.1 Encourage feedback from all levels of employees within the project with regard to problems, ideas, and solutions related to safety;

4.2.6.2 Provide recommendations to the Project Manager, HPAS and the Head Office OSHC regarding health and safety issues;

4.2.6.3 Inform (orient) employees about new safety policies, training Programs, injury risks and causes, and other safety-related matters;

4.2.6.4 Ensure those employees and others (Visitors, Consultants, Subcontractors, etc., as applicable) are informed about safety Policies, training programs, injury risks and causation, and other health and safety-related matters;

4.2.6.5 Identify unsafe work practices and conditions and suggest and/or take appropriate remedies;

4.2.6.6 Assist the HPAS and OSHC in the development and updating of safe operating procedures;

4.2.6.6 Conduct regular (bi-weekly or monthly) safety tours. (See Safety Tour);

4.2.6.7 Review monthly occupational safety and health report of the project;

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4.2.6.8 Hold at least monthly meetings scheduled by the Head,Project Administration Section or the assigned chairperson of the SOSHC.

4.3 General Safety and Health Rules

4.3.1 Every employee shall comply with applicable Government and the Company's safety rules and regulations as a condition of employment.

4.3.2 All injuries, regardless of how minor, must be reported to the immediate or higher supervisor or safety & health officer/assistant and the office immediately. An employee that fails to report an injury immediately will be issued a safety violation notice and may be subject to termination.

4.3.3 Unsafe conditions and near accidents should be reported to supervisors or safety & health officer/assistant so corrective action shall be taken.

4.3.4 Safety glasses will be worn as the minimum required eye protection. Additional eye and face protection such as mono-goggles and face shields are required for such operations as welding, grinding, jack hammering, utilizing compressed air or handling of chemicals, acid and caustic materials. Burning goggles for cutting, burning or brazing and welding hoods for welding are required.

4.3.5 Necessary measures for fall protection should be taken any time where there is a fall hazard.

4.3.6 Clothing must provide adequate protection for the body. Work boots with rigid, slip-resistant soles which give adequate protection to the feet and ankles are required.

4.3.7 Alcoholic beverages or illegal drugs including 'chat' are not allowed at the office or on site. The use or possession of illegal drugs or alcoholic beverages will result in immediate termination.

4.3.8 Housekeeping shall be an integral part of every job. Supervisors and employees are responsible for keeping their work areas clean and hazard free. Clean up is required when finishing a job, at the end of the day or as needed to maintain a safe work place.

4.3.9 Burning and cutting equipment shall be checked daily before being used. All gas shall be shut off and hoses disconnected from bottles or manifolds at the end of each day. Caps shall be replaced on bottles when gauges are removed. All repairs shall be made in the appropriate place at the workshops. Make-shift field repairs

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will not be allowed. Two sets of flashback arresters shall be installed on oxyacetylene outfits; one set at the regulators and one set at the torch handle.

4.3.10 All tools, whether company, personal, or third party (including Subcontractors) must be in good working condition. Defective tools must not be used. Examples are: chisels with mushroomed heads; hammers with loose or split handles; and guards missing on saws or grinders, etc.

4.3.11 The vehicle driving speed should be limited within the construction site. No one is permitted to ride on a truck while standing up. Sitting on the outside edges is prohibited. Riding as a passenger on equipment is prohibited unless the equipment has the safe capability of transporting personnel.

4.3.12 Adequate precautions must be taken to protect employees and equipment from hot work such as welding or burning. Fire extinguishing equipment shall be not further away than 15 meters from all hot work. Empty fire extinguishers shall be returned to the office for refilling immediately.

4.3.13 All scaffolding and work platforms must be in accordance with acceptable specifications. All ladders must be in a safe condition with no broken rungs or split side rails. Damaged ladders shall be removed from service. Ladders shall be secured at the top and bottom. Metal ladders around electrical work are prohibited.

4.3.14 All floor openings or excavations shall be barricaded on all sides to ensure that employees & other people are aware of the hazard. Manholes shall be covered with a secure cover and clearly marked.

4.3.15 Warning signs, barricades, safety posters and tags will be used as much as possible and shall be respected.

4.3.16 Respiratory protection is required for employees exposed to dust hazards or to other air contaminants that may be encountered.

4.3.17 All Project staff should give due attention to take care of themselves, their family and coworkers against environmental diseases caused by vectors such as malaria by keeping their living area clean and dry. In addition to this, everyone has to protect him/herself from mosquito using mosquito net (which may be supplied by the organization for free or at a lower price) or other means of prevention, such as insecticide.

4.3.18 Environmental diseases caused by unclean drinking water are common health hazards associated with project areas. Therefore, the Project Office shall attempt to provide purified or boiled drinking water for the staff at office. And every staff

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should prevent him/herself from these diseases at least by drinking boiled water and avoiding raw (uncooked) food at home.

4.3.19 Safety equipment's such as personal protective equipment's (goggles, face shield, dust mask, gloves, safety belt, helmet) ,warning signs, fire extinguishers, first aid kits etc. required for each project shall be identified, approved and purchased (or produced) during the mobilization period of the project.

4.3.20 A number of the company's employees are highly exposed to natural and manmade hazards that should be mitigated with the joint effort of the organization and themselves. Truck Drivers and Operators are advised to avoid behaviors that expose to unsafe sex and thereby to sexually transmitted diseases including HIV/AIDS. Female workers should be protected from sexual harassment, rape and abuse. The impact of drugs, alcohol and 'chat' on young employees should be well understood and necessary actions should be taken to safeguard from the bad consequence of them.

4.3.21 Workers below age 14 and those who have chronic diseases like epileptic cases which may exposes them for accidents should not permitted to enter and work at site.

4.4 General Work Area (Site) Safety

4.4.1 Employee Training and Orientation

Employees shall be thoroughly trained in the use of protective equipment, guards, and safeguards for chemicals and safe operation of equipment, machines, and tools they use or operate. Only employees who have been trained and those undergoing supervised on-the-job training (OJT) shall be allowed to use equipment, machines, and tools. When new employees recruited, they must be trained & aware about the company general safety & health rules and regulations.

4.4.2 Personal Protective Equipment (PPE)

4.4.2.1 Personal protective equipment (PPE) is not a substitute for safety i.e. engineering and administrative controls or safe working methods and procedures. If it has been determined that control methods are not feasible (enough), personal protective equipment is required whenever there are hazards that can do bodily harm through absorption, inhalation, or physical contact and strike. This equipment includes

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respiratory and hearing protective devices, special clothing, fall protective devices, and protective devices for the eyes, face, head, hand, and extremities.

4.4.2.2 Supervisors shall ensure that personnel use the protective clothing and equipment that will protect them from hazards of the work they perform. It is the responsibility of workers to keep their PPE in a clean, sanitary state of repair and use the equipment when required.

4.4.2.3 PPE should be purchased and distributed for concerned employees when the project starts.

4.4.2.4 Workers shall not wear rings, earrings, bracelets, wristwatches, or necklaces in the vicinity of operating machinery and power tools. Additionally, long full beards, unrestrained long hair, and loose clothing can become caught in tools or machinery and cause serious personal injury. Highly combustible garments or coveralls made of material such as nylon shall not be worn in or around high temperature equipment or operations such as boiler operations, welding, and any other work with open flame devices.

4.4.2.5 Workers shall keep their hands and face clean, change clothes when they are contaminated with solvents, lubricants, or fuels, and keep their hands and soiled objects out of their mouth. No food or drink shall be brought into or consumed in areas exposed to toxic materials or chemicals. Workers shall wash their hands before eating or smoking after exposure to any contaminant.

4.4.3 Safety Notes & Symbols

4.4.3.1 Safety notes & symbols is one of the major safety works that should be done in the construction company. Safety notes & symbols are essential for awareing the employees and indicating that there is something which can harm or cause damage for somebody.

4.4.3.2 Employees should understand and follow the meaning of such symbols & notes by seeing and reading and follow their instruction.

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Employers and concerned bodies are also responsible to tell the meaning of such symbols and notes for the workers.

4.4.3.3 Safety notes & symbols may be produced in the company if possible, prepared by the legal advertising body or may be purchased from the local safety materials shop whenever possible and should be distributed to the projects during the mobilization period.it should be returned to the HO when the project phase out by the concerned body i.e. SHO or HPAS at PO.

4.4.3.4 Safety notes & Symbols should be protected from any damage and used only for the required purpose.

Fig 1. Some common safety notes & symbols used in the construction set up are seen below.



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መንገዱ ዝግነወ.

The road is closed

በመሠራት ላይ ያለ መንገድ

Road under construction

ሰዎች በስራ ላይ

Men Working



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4.5 Work Area Layout

- 4.5.1 Proper layout, spacing, and arrangement of equipment, machinery, passageways, and aisles are essential to orderly operations and to avoid congestion.
- 4.5.2 Equipment and machinery shall be arranged to permit an even flow of materials. Sufficient space should be provided to handle the material with the least possible interference from or to workers or other work being performed. Machines should be placed so it is not necessary for an operator to stand in a passageway. Additionally, machine positioning should allow for easy maintenance, cleaning, and removal of scrap.
- 4.5.3 Machines (e.g. generators, mixers, crusher engines) with shock mounting pads shall be securely anchored and installed according to manufacturer's instructions.
- 4.5.4 Passage ways/aisles shall be provided and marked to permit the free movement of employees bringing and removing material from the work area. These passageways are independent of clear zones and storage spaces. They shall be clearly recognizable.
- 4.5.5 Where powered materials handling (loading, unloading and transporting) equipment such as, forklift, loader or dumper, is used, facility layout shall provide enough clearance in passageways, on loading docks and through doorways to permit safe turns. Aisles shall be at least 1 meter wider than the widest vehicle used or most common material being transported.

4.6 Illumination and Ventilation

- 4.6.1 Adequate illumination (light) shall be provided to ensure safe working conditions in office, store, and workshops, and when working out doors (at site) on night or when natural illumination is weak or inadequate. Avoiding or minimizing excessive illumination or reflected glare (light reflections) is also crucial to control hazards related to excess light exposure.
- 4.6.2 There should be satisfactory ventilation for the work place, i.e. (a) the provision of sufficient air for people to breathe and work in reasonable comfort and, (b) means of removal of air which may become suffocated or contaminated as a result of process.

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4.7 Exits and Exit Markings

- 4.7.1 Exit access shall be arranged so it is unnecessary to travel toward any area of high hazard potential in order to reach the nearest exit (unless the path of travel is effectively shielded by suitable partitions or other physical barriers). Areas around exit doors and passageways shall be free of obstructions.
- 4.7.2 Where the blocking of any single exit due to fire or smoke may endanger occupants, there shall be at least two means of exit remote from each other.
- 4.7.3 Exits, exterior steps, and ramps shall be adequately lighted to prevent mishaps. Separate lighting will not be required if street or other permanent lighting gives at least one foot-candle of illumination on the exit, steps, or ramp.

4.8 House Keeping and Access at Site

Good housekeeping shall be maintained in all workshops, yards, buildings, and mobile equipment. Supervisors are responsible for good housekeeping in or around the work they are supervising. As a minimum, the following requirements shall be adhered to:

- 4.8.1 Material shall not be placed where anyone might stumble over it, where it might fall on someone, or on against any support unless the support can withstand the additional weight.
- 4.8.2 Aisles and passageways shall be kept clear of tripping hazards.
- 4.8.3 Nails shall be removed from loose lumber or the points turned down.
- 4.8.4 Trash and other waste materials shall be kept in approved receptacles. Trash shall not be allowed to accumulate and shall be removed and disposed of as soon as possible.
- 4.8.5 The floor of every work area shall be maintained so far as practicable, in a dry condition. Where wet processes are used, drainage shall be maintained and removable false floors, platforms, mats, or other dry standing places be provided.
- 4.8.6 Distribution panels, fire extinguishers, switch boards, or alarm supply boxes shall not be blocked by any obstruction, which may prevent ready access.
- 4.8.7 Machinery and equipment shall be kept clean of excess grease, oil and excessive dust. Pressure gauges and visual displays shall be kept clean, visible, and

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serviceable at all times. Drip pans and wheeled or stationary containers shall be cleaned and emptied at the end of each operation.

4.9 Fire Prevention

4.9.1 All engineering personnel, store keepers, guards, mechanics, Drivers and Machine Operators shall receive fire prevention training as part of their general training.

4.9.2 Supervisors in charge of operations where fuels, solvents, or other flammable liquids and/or fire are used shall be constantly alert for hazards and unsafe acts and ensure that leaks or spills of flammable or combustible materials are leaned up promptly.

4.9.3 Fuels such as gasoline shall never be used to clean floors or clothing, and open solvent or gasoline containers shall not be kept near electrical equipment. Open flames, open element heaters, equipment not properly grounded, and non-explosion-proof electrical equipment used in the presence of flammable or combustible liquids shall be avoided.

4.9.4 Fire extinguishers shall be installed in work areas exposed to fire hazard. The number of extinguishers depends upon the size and layout of the facility. Fire extinguishers shall meet the following requirements:

4.9.4.1 Be kept fully charged and in their designated area.

4.9.4.2 Be located along normal paths of travel.

4.9.4.3 Not be obstructed or obscured from view.

4.9.4.4 Be visually inspected at least monthly to ensure that they:

a. Are in their designated places.

b. Have not been tampered with or actuated.

c. Do not have corrosion or other impairments.

d. Are accessible and not obstructed.

4.9.4.5 Be examined at least yearly and/or recharged or repaired to ensure operability and safety.

4.9.4.6 Be hydrostatically tested.

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4.9.4.7 Be placed so the maximum travel distance, unless there are extremely hazardous conditions, does not exceed 20 meter feet for Class A and Class D or 15 meter for Class B and Class C locations.

Table1.Fire and fire appliance classification:

Class of fire	Description of fire	Appropriate extinguisher
A	Involves solid materials such as wood, paper and cloth which produce glowing embers (coals) or chars.	Water, foam, dry powder, vaporizing liquid, carbon dioxide
B	Involves flammable gases, liquids, and greases, including gasoline and most hydrocarbon liquids which must be vaporized for combustion to occur.	Foam, carbon dioxide, dry powder, and water - only for fires involving water soluble materials such as <i>methanol</i> and <i>acetone</i>
C	Involves fires in live electrical equipment or in materials near electrically powered equipment	Carbon dioxide and dry powder
D	Involve combustible metals, such as magnesium, zirconium, potassium, and sodium.	Carbon dioxide and dry powder

4.9.4.8 Water supply line with plastic hose shall be made available and ready in work areas exposed Class-A fire hazard.

4.9.4.9 Supervisors shall ensure that employees remove construction debris and rubbish from the job site upon completion of the job, or daily if extended beyond one day. Hazardous materials shall not be left at job sites unless properly stored.

4.9.4.10 Spraying of paint, solvents, or other types of flammable materials should be avoided in rooms with poor ventilation, as build-up of fumes and vapors can cause explosions or fires.

4.9.4.11 LP gas tanks shouldn't be stored inside buildings.

4.9.4.12 Temporary heaters should be kept at least 2 meter away from any LP gas container.

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4.10 Excavation and Trenching

4.10.1 General

During Excavation and Trenching, all the concerned:-

- 4.10.1.1 Should find the location of all underground utilities by contacting the local utility locating service before digging.
- 4.10.1.2 Should keep workers away from digging equipment and never allow workers in an excavation when equipment is in use.
- 4.10.1.3 Should keep workers from getting between equipment in use and other obstacles and machinery that can cause crushing hazards.
- 4.10.1.4 Should keep equipment and the excavated dirt (spoils pile) back 60 cm from the edge of the excavation.
- 4.10.1.5 Should have a competent person conduct daily inspections and correct any hazards before workers enter a trench or excavation.
- 4.10.1.6 Should provide workers a way to get into and out of a trench or excavation such as ladders and ramps. They must be within 75 cm of the worker.
- 4.10.1.7 Should use equipment's for excavations of utility trenches over 1.5 meter deep, use shoring, shields (trench boxes), benching, or slope back the sides. Unless soil analysis has been completed, the earth's slope should be at least 1½ meter horizontal to 1 meter vertical.
- 4.10.1.8 Should keep water out of trenches with a pump or drainage system, and inspect the area for soil movement and potential cave-ins.
- 4.10.1.9 Should keep drivers in the cab and workers away from dump trucks when dirt and other debris are being loaded into them. Workers shouldn't allow under any load and should be trained to stay clear of the backs of vehicles.
- 4.10.1.10 During the excavation work, all open holes should be guarded to prevent employees and other people to get accidentally in to the holes at night or day time.

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4.10.2 Foundations

4.10.2.1 After the foundation walls are constructed, take special precautions to prevent injury from cave-ins in the area between the excavation wall and the foundation wall.

4.10.2.2 The depth of the foundation/basement trench shouldn't exceed 2.3 meter deep unless you provide other cave-in protection.

4.10.2.3 The horizontal width of the foundation trench should be kept at least 60 cm wide.

4.10.2.4 No work activity should vibrate the soil while workers are in the trench.

4.10.2.5 The foundation trench work should be planned to minimize the number of workers in the trench and the length of time they spend there.

4.10.2.6 The trench should be inspected regularly for changes in the stability of the earth (water, cracks, vibrations, spoils pile). Work should be stopped if any potential for cave-in develops and fix the problem before work starts again.

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4.11 Tools and Equipment Safety

4.11.1 General

4.11.1.1 Employees should be trained in the proper use of all tools and equipment's. Workers should be able to recognize the hazards associated with the different types of tools and the safety precautions necessary.

4.11.1.2 All hand and power tools should be maintained in a safe condition and checked regularly for defects. Broken or damaged tools and equipment's should be removed from the jobsite.

4.11.1.3 Examine each tool for damage before use and do not use damaged tools.

4.11.1.4 The right tool should be used for right job. And the manufacturer's requirements and instruction should be followed for safe use of all tools.

4.11.1.5 Appropriate personal protective equipment such as safety goggles and gloves must be worn to protect against hazards that may be encountered while using hand tools.

4.11.2 Hand Tools

Hand tools are tools that are powered manually. Incidents at the job site involving hand tools are usually the result of misuse. Hand tools are precision tools capable of performing many jobs when used properly. Prevention of incidents involving hand tools on the job site becomes a matter of good instruction, adequate training, and proper use. The greatest hazards posed by hand tools result from misuse and improper maintenance. Some examples include the following:

4.11.2.1 If a chisel is used as a screwdriver, the tip of the chisel may break and fly off, hitting the user or other employees.

4.11.2.2 If a wooden handle on a tool, such as a hammer or an axe, is loose splintered, or cracked, the head of the tool may fly off and strike the user or other employees.

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4.11.2.3 If the jaws of a wrench are sprung, the wrench might slip.

4.11.2.4 If impact tools such as chisels, wedges, or drift pins etc. have mushroomed heads, the heads might shatter on impact, sending sharp fragments flying toward the user or other employees.

To prevent hazards associated with the use of hand tools employees should observe the following precautions.

4.11.2.5 The supervisors as well as the employees are responsible for the safe condition of tools and equipment used by employees.

4.11.2.6 Supervisors shall not issue or permit the use of unsafe hand tools.

4.11.2.7 Employees, when using saw blades, knives, or other tools, should direct the tools away from aisle areas and away from other employees working in close proximity.

4.11.2.8 Knives and scissors must be sharp; dull tools can cause more hazards than sharp ones.

4.11.2.9 Cracked saw blades must be removed from service.

4.11.2.10 Wrenches must not be used when jaws are sprung to the point that slippage occurs.

4.11.2.11 Impact tools such as drift pins, wedges, and chisels must be kept free of mushroomed heads.

4.11.2.12 Wooden handles should be kept free of splinters or cracks and be sure the handles stay tight in the tool.

4.11.2.13 Workplace floors shall be kept as clean and dry as possible to

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prevent accidental slips with or around dangerous hand tools.

4.11.1 Power Tools

Portable power tools increase mobility and convenience but are frequently more hazardous to use than their stationary counterparts. Power tools must be fitted with guards and safety switches; they are extremely hazardous when used improperly. The types of power tools are determined by their power source:

4.11.3.1 Electric power tools: Electric power tools are operating using .electric power. E.g. girders, drills, power saw, abrasive wheels

4.11.3.2 Pneumatic power tools: Pneumatic power tools are powered by compressed air and include chippers, drills, jack hammers, and sanders.

4.11.3.3 Liquid fuel power tools: Fuel-powered tools are usually operated with gasoline. E.g. power saw, hand compressors.

4.11.3.4 Hydraulic power tools: Hydraulic power tools are tools powered by fire resistant fluid. E.g. All jacks-including lever and ratchet, jacks, *screw jacks*.

To prevent hazards associated with the use of power tools, workers

Should observe the following general precautions:

4.11.3.5 Personnel who are required to use portable power tools in their work shall be thoroughly trained in safe operating practices prior to using the tools.

4.11.3.6 Safe operating procedure shall be set up for each type of tool Consistent with the manufacturer's instructions.

4.11.3.7 A tool shall never been carried by the cord or hose and never yank the cord or the hose to disconnect it from the power outlet.

4.11.3.8 Cords and hoses should be kept away from heat, oil, and sharp edges.

4.11.3.9 Tools should be disconnected when not using them, before servicing and cleaning them, as well as when changing accessories such as blades, bits, and cutters.

4.11.3.10 All people that are not involved with the work should be kept at a safe distance from the work area.

4.11.3.11 The work should be secured with clamps or a vise, so as to free both hands to operate the tool.

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- 4.11.3.12 Accidental starting should be avoided. Fingers should not hold on the switch button while carrying a plugged-in tool.
- 4.11.3.13 Maintain tools with care; keep them sharp and clean for best performance. Follow instructions in the user's manual for lubricating and changing accessories.
- 4.11.3.14 everyone should be sure to keep good footing and maintain good balance when operating power tools.
- 4.11.3.15 Proper clothes should be used for the task. Loose clothing, ties, or jewelry can become caught in moving parts, such as grinding wheel and circular saw blade.
- 4.11.3.16 All damaged portable power tools should be removed from use and should be tagged: "Do Not Use."
- 4.11.3.17 Double insulated electric tools should be used or the tools should be grounded.
- 4.11.3.18 Cords from electric tools shouldn't present a tripping hazard.
- 4.11.3.19 Electric tools should be stored in a dry place when not in use. They shouldn't be used in damp or wet locations unless they are approved for that purpose.
- 4.11.3.20 The exposed moving parts of power tools need to be safeguarded. Belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, or other reciprocating, rotating, or moving parts of equipment must be guarded. Safety guards must never be removed when a power tool is being used.
- 4.11.3.21 Power tools should be turned off before leaving them unattended. Equipment should be kept in a safe place, according to the manufacturer's instructions.
- 4.11.3.22 Hose connections shall always be secured when using pneumatic tools.
- 4.11.3.23 Compressed air guns should never be pointed toward anyone. Workers should never "dead-end" them against themselves or anyone else.
- 4.11.3.24 Eye protection should be required, and head and face protection should be recommended for employees working with pneumatic tools. For example, workers operating a jackhammer must wear safety glasses and safety shoes that protect them against injury if the jackhammer slips or falls. A face shield also should be used. Noise is another hazard associated with pneumatic tools. Working with noisy tools such as jackhammers requires proper and effective use of appropriate hearing protection.

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4.11.3.25 All jacks including lever and ratchet jacks, screw jacks, and hydraulic jacks must have a stop indicator, and the stop limit must not be exceeded. Also, the manufacturer's load limit must be permanently marked in a prominent place on the jack, and the load limit must not be exceeded.

4.11.3.26 A jack should never be used to support a lifted load. Once the load has been lifted, it must immediately be blocked up. A block should be put under the base of the jack when the foundation is not firm, and a block should be placed between the jack caps and load if the cap might slip.

4.11.3.27 To set up a jack, that the base of the jack should rest on a firm, level surface; the jack should be correctly centered; the jack head should bear against a level surface; and the lift force should be applied evenly.

4.11.3.28 As proper maintenance of jacks is essential for safety, all jacks must be lubricated and inspected regularly.

4.11.4 Vehicles and Mobile Equipment safety

In order to protect the safety of passengers and property, operators, drivers and passengers shall take the necessary precautions and adhere to safety rules stipulated by traffic regulations, the transport policy and procedures and these policies and procedures.

The following shall be respected, among others, by all concerned.

4.11.4.1 Workers should be trained to stay clear of backing and turning Vehicles and equipment with rotating cabs.

4.11.4.2 All off-road equipment used on site should be equipped with Rollover Protections (ROPs).

4.11.4.3 Back-up alarms should be maintained for equipment with limited rear view or use someone to help guide them back.

4.11.4.4 All vehicles shall have fully operational braking systems and brake lights.

4.11.4.5 Seat belts should be used when transporting workers in motor and construction vehicles.

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- 4.11.4.6 At least a 3-meter clearance from overhead power lines should be maintained when operating equipment.
- 4.11.4.7 The raised bed should be blocked up when inspecting or repairing dump trucks.
- 4.11.4.8 The rated capacity of the crane should be known and used accordingly, and it should be ensured that the crane is stable.
- 4.11.4.9 A tag line should be used to control materials moved by a crane.
- 4.11.4.10 Experience should be verified or training should be provided to crane and heavy equipment operators.
- 4.11.4.11 Drivers and passengers should wear seat belts specially while driving out of town.
- 4.11.4.12 Truck drivers and heavy duty machine operators should always have an assistant.
- 4.11.4.13 Driving or operating under the influence of alcohol, chat or intoxicating drugs is prohibited.
- 4.11.4.14 Before starting to drive or operate, the driver or the operator should ensure that the vehicle is checked for water, oil and fuel and that it is in good mechanical order.
- 4.11.4.15 Speed of driving should be kept within acceptable limits suitable to the traffic, type of road, safety of passengers, etc. As a general rule, the maximum speed on highways outside cities shall not exceed 100 km/hr and in built-up areas and villages 40km/hr.
- 4.11.4.16 All vehicles and heavy machineries shall have fire extinguishers.
- 4.11.4.17 Vehicles and machineries should not carry passengers or goods over their capacity. Objects obstructing proper vision should be avoided, Materials should be loaded properly to avoid damages to the materials as well as the vehicle.
- 4.11.4.18 Drivers should not allow unauthorized persons or load to travel in their vehicles.
- 4.11.4.19 Drivers should have a valid driving license and should be familiar with traffic rules and regulations.(Fines due to traffic offense shall be borne by the driver)
- 4.11.4.20 Lights should be on while driving in heavy rains, dusty roads and foggy weather.
- 4.11.4.21 Four-wheel drives should be used for off-road driving.
- 4.11.4.22 Driving while sick or sleepy should be avoided.
- 4.11.4.23 Night driving should be avoided when situations might be dangerous.

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4.11.4.24 To ensure that vehicles and machineries are secured, they shall be parked in secured areas; project sites compounds, hotel compounds, etc.

4.11.4.25 In cases of breakdown on roads, someone should be assigned to guard the vehicle until removal.

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4.12 Material Storage and Handling

4.12.1 All unnecessary accumulation of materials and supplies in the workshops/work area shall be avoided. The presence of unnecessary material in the workshops/work area could cause such incidents as tripping, falling, or slipping. This could be especially hazardous around equipment that is in operation. The only material in the workshops area shall be that actually in work. The only places that materials should accumulate in quantity are in storerooms and material holding areas.

4.12.2 The storage of materials shall not, of itself, create a hazard. Materials stored in rows shall be stacked, strapped, blocked or interlocked, and limited in height so they are stable and secure against sliding or collapse. Storage racks shall have sufficient capacity to bear the loads imposed on them.

4.12.3 Stored materials shall not obstruct fire extinguishers, alarm boxes, sprinkler system controls, electrical switch boxes, machine operations, emergency lighting, first aid or emergency equipment, or exits.

4.12.4 Heavy materials and equipment should be stored low and close to the ground or floor to reduce the possibility of injury during handling.

4.12.5 All passageways and storerooms shall be maintained clean, unobstructed, dry, and in sanitary condition. Spills will be promptly removed.

4.12.6 Where mechanical handling equipment, such as lift trucks and dumpers are used, safety clearance shall be provided for aisles at loading docks, through doorways, and wherever turns or passages must be made. No obstructions that could create a hazard are permitted in aisles.

4.12.7 Gasoline and other flammable liquids shall be stored in a safety can outdoors or in an approved storage facility. And any leaks or spills of flammable or combustible materials shall be cleaned up promptly

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4.13 Working at Elevations

These procedures are designed to prevent the injury of The Company employee due to falls or slips any time personnel are working on portable ladders, or scaffolding, or at elevations of more than 1.5 meter above grade.

4.13.1 Ladders

- 4.13.1.1 Falls are the primary hazard associated with the use of ladders. Falls result from a number of unsafe acts and conditions such as: (1) Ladders being set on unstable surfaces. (2) Personnel reaching too far out to the sides. (3) Personnel standing too high to maintain balance. (4) Personnel using defective ladders (e.g., broken rails, rungs, missing hardware).
- 4.13.1.2 These hazards are minimized if workers adhere to proper ladder safety practices and if supervisors ensure equipment is used, inspected, and maintained in good condition. Workers capable of the physical exertion required under these conditions must accomplish tasks, which require frequent use of ladders and involve significant climbing effort.
- 4.13.1.3 Wooden parts used in construction of ladders should be straight-grained; thoroughly seasoned; smoothly dressed; and free of sharp edges, splinters, checks, decay and other defects. Rungs must be parallel, level and uniformly spaced. The spacing shall not be more than 12 inches.
- 4.13.1.4 Personnel shall not use portable metal ladders when performing work on or near electrical equipment. Wood or reinforced plastic ladders shall be used for work on or near electrical equipment. They will be kept clean. All surface buildup or dirt, grease, or oils should be removed to avoid creating a ready path for electrical current.
- 4.13.1.5 Where possible, portable non self-supporting ladders will be used at such a pitch that the base of the ladder is placed a distance from the vertical wall that is one-fourth of the working length of the ladder. The ladder shall be placed to prevent slipping, or it will be lashed or manually held in position.
- 4.13.1.6 More than one person shall not use ladders at a time. Ladders specially designed to support greater loads shall be used in combination with

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ladder jacks and scaffold planks when an operation requires more than one person.

- 4.13.1.7 Ladders shall not be placed in front of doors opening toward the ladder unless the door is blocked open, locked, or guarded.
- 4.13.1.8 Ladders should not be placed on boxes, barrels, or other unstable bases to obtain additional height.
- 4.13.1.9 To support the top of the ladder at a window opening, attach a board across the back of the ladder, extending across the window to provide firm support against the building walls or window frames.
- 4.13.1.10 When ascending or descending, users shall face the ladder and use both hands.
- 4.13.1.11 Ladders with broken or missing steps, rungs, or cleats, broken side rails, or other defects shall not be used. Improvised repairs should not be used as well.
- 4.13.1.12 Non-slip bases should not be intended as a substitute for care in safely placing, lashing, or holding a ladder that is being used upon oily, metal, concrete, or slippery surfaces.
- 4.13.1.13 When the ladder can be knocked over by others who are working in the area, the ladder will be securely fastened. As an alternative, someone will be assigned to steady the bottom, or the area around the ladder will be roped off.

4.13.2 Scaffolding and Elevated Platform

In order to minimize the dangers and hazards related to scaffolding and elevated platform works and all the concerned:-

- 4.13.2.1 Should provide safe access to get on and off scaffolds and work platforms safely.
- 4.13.2.2 Shouldkeep scaffolds and work platforms free of debris. Also, tools and materials should be kept as neat as possible on scaffolds and platforms. These practices will help prevent materials from falling and workers from tripping.

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- 4.13.2.3 Should erect scaffolds on firm and level foundations. Finished floors will normally support the load for a scaffold or work platform and provide a stable base.
- 4.13.2.4 Should place scaffold legs on firm footing and secure from movement or tipping, especially surfaces on dirt or similar surfaces.
- 4.13.2.5 Should erect and dismantle scaffolds only under the supervision of a competent person.
- 4.13.2.6 Should ensure that each scaffold is capable of supporting its own weight and 4 times the maximum intended load.
- 4.13.2.7 Should use manufactured base plates or mudsills made of hardwood or equivalent to level or stabilize the footings. Don't use blocks, bricks, or pieces of lumber.
- 4.13.2.8 Should assign a competent person to inspect scaffolds before each use.
- 4.13.2.9 Should guard scaffold platforms that are more than 3 meter above the ground or floor surface with a standard guardrail. If guardrails are not practical, the use of other fall protection devices such as safety harnesses and lanyards is advised.
- 4.13.2.10 Shouldn't allow employees to work on scaffolds when they are feeling weak, sick, or dizzy.
- 4.13.2.11 Shouldn't move a scaffold horizontally while workers are on it, unless it is a mobile scaffold and the proper procedures are followed.
- 4.13.2.12 Shouldn't allow employees to work on scaffolds covered with slippery materials.
- 4.13.2.13 Shouldn't erect, use, alter, or move scaffolds within 3 meter of overhead power lines.
- 4.13.2.14 Shouldn't work on scaffolds in bad weather or high winds unless the competent person decides that doing so is safe.
- 4.13.2.15 Shouldn't use ladders, boxes, barrels, or other makeshift contraptions to raise your work height.

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4.13.2.16 Shouldn't let extra material build up on the platforms.

4.13.3 Fall Protection

As a Fall Protection:-

4.13.3.1 Guardrails should be installed around openings in floors and across openings in walls when the fall distance is 3 meter or more. It should also be ensured that the top rails can withstand a 100 kg load.

4.13.3.2 Guardrails should also be constructed with a top rail approximately 1 meter high with a mid-rail about half that high at 50 cm.

4.13.3.3 Toe boards should also be installed when other workers are to be below the work area.

4.13.3.4 Floor openings larger than 5x5 cm should be covered with material to safely support the working load.

4.13.3.5 Other fall protection systems such as slide guards, roof anchors or alternative safe work practices should be used when a guardrail system cannot be used.

4.13.3.6 Proper slip-resistant shoes or footwear should be worn to lessen slipping hazards.

4.13.3.7 Workers should be trained in safe work practices before performing work on foundation walls, roofs, trusses, or before they perform exterior wall erections and floor installations.

4.13.4 Precautions against falling objects

4.13.4.1 Falls should be prevented from height by physical precautions or, where this is not possible, provide equipment that will arrest falls (e.g., net).

4.13.4.2 Materials or objects should not be thrown from height if they could probably strike some one.

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4.14 Welding and Metal fabrication Safety

The following provides minimum guidance on operating precautions and procedures.

- 4.14.1 Before lighting the torch for the first time each day, enough of each gas should be allowed to flow through its respective hose to purge any flammable gas mixture. Purge hoses in open spaces and away from ignition sources. The torch should be lightened with a friction lighter or stationary pilot flame keeping a safe distance between the torch and the welder's hands. The torch should be pointed away from persons or combustible materials when lighting. Attempt shouldn't be made to light a torch from hot metal.
- 4.14.2 Fuel gas (acetylene) and oxygen torch valves shall be closed and the fuel gas and oxygen supply to the torch shall be shut off during lunch or break periods, when not in use for extended periods, and when unattended.
- 4.14.3 Welding torch hoses must be protected from damage by contact with hot metal, open flames, corrosive agents or sharp edges. Pressure on hoses will be released at the end of each workday. Hoses must be visually inspected for damage at the beginning of each shift. Hose showing leaks, cuts, burns, worn spots or other evidence of deterioration must be repaired or replaced prior to use. Replacement hoses or fittings must be approved for use with acetylene equipment.
- 4.14.4 A fire watch will be maintained for at least 30 minutes after completion of cutting or welding operations to detect and extinguish possible smoldering fires.
- 4.14.5 Goggles or other suitable eye protection should also be worn during arc welding or cutting and all gas welding or oxygen cutting operations to provide protection from injurious rays from adjacent work, and from flying objects.
- 4.14.6 All welders should not wear fire sensitive aprons, and coveralls (made of nylon & polyester etc.).
- 4.14.7 Clothing should be kept reasonably free of oil or grease. Front pockets and upturned sleeves or cuffs should be prohibited, and sleeves and collars should be kept buttoned to prevent hot metal slag or sparks from contacting the skin. The welding operation environment shall be free of flammable liquids and vapors. Combustible materials within a radius of 35 feet of the operation will be protected from activity residue (flame, heat, sparks, slag, etc.).
- 4.14.8 Fire extinguishing equipment shall be maintained, ready for use, while welding or cutting operations are being performed.

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4.14.9 Before cutting, welding, or performing any hot work on any tank (e.g. fuel tank of vehicles), cylinder, or other container, the item shall be purged or made inert with water or air. New containers shall also be made inert as they may contain a flammable preservative, which could form explosive vapors when heated. Welders shall also ensure that there are no substances such as grease, tars, or acids which, when subjected to heat, might produce explosive or toxic vapors. Any pipelines or connections to the drums, cylinders, tanks, or other containers shall be disconnected or blanked.

4.14.10 Arc welding equipment shall conform to the design and installation criteria of acceptable standard "Welding, Cutting, and Brazing." The frame or case of the welding machine (except engine-driven machines) shall be grounded under the conditions and according to the methods prescribed in the operating and users' manual.

4.14.11 Before starting operations, all connections to the arc-welding machine shall be checked. Coiled welding cable shall be spread out before use to avoid serious overheating and damage to insulation. Power supply and welding cables shall be inspected for damage and wear before use. Cables with damaged insulation or exposed conductors shall be replaced. Electrode cables shall be joined and insulated in accordance with approved methods.

4.14.12 Grounding of the welding machine shall be checked. Special attention shall be given to the ground connections of portable machines.

4.14.13 Electrode holders, when not in use, shall be placed where they cannot make electrical contact with persons, conducting objects, fuel, or compressed gas cylinders.

4.14.14 When it is necessary to splice cables to extend their length, only certified electricians shall make the splices. Cables with splices within 3 meter of the electrode holder shall not be used. The welder shall not coil or loop welding electrode cables around parts of their body.

4.14.15 Welders shall not place welding cable and other equipment where it will obstruct passageways, ladders, and stairways.

4.14.16 Welding machines, which have become wet shall be thoroughly dried and tested before being used.

4.14.17 When welders are working close to one another on one structure where they may touch the exposed parts of more than one electrode holder simultaneously, the machines shall be connected to minimize shock hazard as follows:

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4.14.17.1 All direct current (DC) machines shall be connected with the same polarity.

4.14.17.2 All alternating current (AC) machines shall be connected to the same phase of the supply circuit and with the same instantaneous polarity.

4.15 Noise prevention

Noise is sound which is unpleasant and not desired. In the construction area there is noise which is mainly emitted from machines used for different purposes. Noise affects our health both physically and psychologically. Physically it causes temporary and permanent hearing loss while psychologically it causes psychological annoyance and stress. Standards of noise exposure vary from country to country. The amount of noise a worker can be exposed varies with its intensity. A worker can be exposed to a low level noise indefinitely, but as the intensity increases above 90dB (A), the period of exposure has to be decreased in order that cumulative hearing damage does not occur. Generally to reduce the effects of noise we focus on the prevention at three areas i.e. at the source, at the path and at the worker.

At the source-By enclosing the entire machine or noisy parts of the machine, by servicing the machine, by substituting the noisy part of the machine with sound absorbent material etc.

At the path-By increasing the distance between the noise source and the worker, placing sound absorbent material between the noise source and the worker etc.

At the worker-At this area noise exposure is decreased by giving different hearing protectors like ear plug, ear muff etc. by reducing exposure time for the worker and so on.

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4.16 Electrical Safety

- 4.16.1** Maintaining or servicing any new and existing energized (hot) electrical circuits shall be prohibited until all power is shut off and de-energized.
- 4.16.2** Frayed or worn electrical cords or cables shall not be used.
- 4.16.3** Only 3-wire type extension cords should be used that are designed for hard or junior hard service.
- 4.16.4** All electrical tools and equipment shall be maintained in safe condition and check regularly for defects.
- 4.16.5** Broken or damaged electrical tools and equipment should be removed from the jobsite.
- 4.16.6** All temporary power (including extension cords) should be protected with Ground Fault Circuit Interrupters (GFCIs). Plug into a GFCI-protected temporary power pole, a GFCI protected generator, or use a GFCI extension cord to protect against shocks.
- 4.16.7** Any protective system or device designed to protect employees from contact with electrical current shouldn't be bypassed.
- 4.16.8** Overhead electrical power lines should be located and identified. Ladders, scaffolds, equipment or materials shouldn't come within 3 meter of electrical power lines.

4.17 Chemical Safety

Chemical have become a part of our life, Sustaining many of our activities. However one cannot ignore that these chemicals may, especially not properly used, endanger our health and poison our environment.

Today, invirtually every work place, workers are exposed to chemicals. The construction Industry also uses different chemicals for different purposes. Many of these chemicals however may endanger the lives of the worker and may pollute the environment. So, to protect from the risks of these chemicals one may consider the following points,

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- 4.17.1 Before working with the chemical one should know about the properties of the chemical, by seeing & reading the chemical data sheet and also by asking the nearby supervisor.
- 4.17.2 To protect the skin, protective equipment's such as gloves, aprons, boots and coveralls made from impervious materials, should be provided to eliminate prolonged contact with solvents or other chemicals. Also to protect the eyes & entering into the inside part of the body one should wear appropriate personal protective equipment like goggles and respirator or mouth mask.
- 4.17.3 After finishing work and before eating or drinking one should thoroughly wash exposed parts of the body with the available detergent or soap.
- 4.17.4 Eating or drinking around the chemical area should be prohibited.
- 4.17.5 One should follow appropriate procedure during handling, transporting storing or disposing of any chemical.
- 4.17.6 Emergency Showers and eye wash points should be provided in close proximity to work stations where hazardous chemicals are used.

Fig2. Common chemical safety signs & symbols



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4.18 Office Safety

Most people would consider the average office to be a reasonably safe workplace compared, say, with a construction site or foundry. However, while fatal accidents are uncommon, minor accidents caused by trips and falls, the unsafe use of electricity, obstructed passages and stairways, failure to use access equipment and poor housekeeping are common.

4.18.1 Housekeeping

One of the greatest causes of accidents in offices, particularly falls, is that of bad housekeeping. The term ‘housekeeping’ implies ‘everything in the correct place and a place for everything’. Bad housekeeping is frequently the cause of office fires, which can be caused by general untidiness, poor storage of flammable wastes, such as waste paper, and people smoking.

The following housekeeping rules should be applied in office.

- 4.18.1.1 The work area should be kept tidy. Items that are not in use should be stored away.
- 4.18.1.2 Waste should be stored in waste containers, which should be emptied on a regular daily basis.
- 4.18.1.3 Heavy items, such as box files, should not be placed on the top of cabinets or cupboards as they could fall on to someone using the cupboard.
- 4.18.1.4 Harmful items, such as broken bottles and light bulbs should be separately stored and disposed off.
- 4.18.1.5 Passages, stairways, entrances and exits, in particular emergency exits of offices should be kept free from surplus stationery materials, surplus office equipment and furniture, and other large items.
- 4.18.1.6 Spillages of liquids should be cleared up immediately to prevent slipping hazards.
- 4.18.1.7 Damaged floor covering, such as carpets, should be replaced immediately to avoid tripping or slipping hazard created.

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4.18.2 Furniture and Fitting

- 4.18.2.1 Bad office planning and layout with regard to placement of furniture and fitting can result in numerous minor accidents. The following points should be considered with regard to office layout and the location of items such as filing cabinets, desks and equipment.
- 4.18.2.2 Furniture should be arranged so that employees can move freely within the office and from one office to another.
- 4.18.2.3 Doors and drawers, particularly to filing cabinets, should be kept closed when not in use. When using a filing cabinet, only one drawer should be opened at a time due to the risk of the cabinet tipping forward, particularly when the two top drawers are open at the same time.
- 4.18.2.4 Filing cabinets should not be overloaded. Heavy items should be stored in the bottom drawer. The bottom drawer of a filing cabinet should always be closed immediately to avoid the risk of people tripping over it.
- 4.18.2.5 Damaged and broken furniture and fitting, such as shelves, chairs and filing cabinets, should be replaced immediately.

4.18.3 Electrical Appliance

- 4.18.3.1 Misuse and abuse of electricity in offices is one of the most significant causes of office fires. There should be a total prohibition on staff undertaking electrical repairs or modification. The increased use of electrical appliances in offices has also contributed to the risk of electrical overloading.

The following rules should be observed.

- a) Only trained and competent staff should attempt to repair electrically operated machinery.
- b) Machines should be switched off from the mains when left unattended for long periods.
- c) Cables should be so positioned that do not trip people up. Where this is not possible, suitable permanent cable covers should be installed. Flexes should be shortened so that they do not trail across the floor or under desks.

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- d) Cables to electrical appliances should be maintained in a sound condition and replaced when they become frayed or damaged.
- e) The use of multi-point adaptors should be either prohibited or carefully controlled in order to avoid overloading sockets.
- f) All electric driven office equipment such as Photocopier, Printer, etc. should be supplied with voltage stabilizers (power regulators). In addition, Uninterruptible Power Supply (UPS) should be used for all computers.

4.18.4 Dangerous Substances

- 4.18.4.1 A variety of products used in offices on a daily basis can be potentially dangerous. Such products include cleaning fluids, adhesives, and quick drying inks and correcting fluids, all of which emit powerful fumes or vapors. To reduce the risks associated with dangerous substances, the following precautions are necessary.
- 4.18.4.2 Staff should always read the manufacturer's instruction prior to using a potentially dangerous product.
- 4.18.4.3 In certain situations it may be necessary to wear personal protective equipment, such as gloves, apron and goggles, when dealing with substances.
- 4.18.4.4 Staff should be aware of the hazard warning symbols shown on the packages for potentially dangerous substances, e.g. flammable, toxic symbols.
- 4.18.4.5 Any ill effects experienced by staff following the use of substances should be reported immediately.

4.18.5 Office Equipment

Certain item of office equipment, in particular hand-operated paper cutter, can inflict serious injury. Paper cutters should be effectively guarded and staff trained in their safe use. Other items, such as scissors, letter openers and knives, should only be used for their main purpose and certainly not, for example, as screwdrivers or for opening tins.

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4.18.6 Fire Precautions

The majority of fires in offices occur outside normal working hours. The initial cause of the fire will, in many cases, have been created during office hours. To minimize the risk of fire, the following procedures and practices should be followed.

- 4.18.6.1 Flammable wastes should be carefully controlled in terms of storage and removal.
- 4.18.6.2 Smoking in the office area should not be allowed. Persons who are careless in their smoking habits should be disciplined.
- 4.18.6.3 All electrical equipment should be disconnected from the socket when in use or left unattended for long periods.
- 4.18.6.4 Every staff, prior to locking the office at the termination of work should undertake an inspection to ensure that no fire hazards have been created.
- 4.18.6.5 Flammable items should be kept to a minimum for general use. A metal storage cupboard should be provided for storage.
- 4.18.6.6 Fire extinguishers and water hoses should be available in appropriate places. Office staff and guards shall be trained on the use of fire extinguishers.

4.18.7 Personal Conduct

Many office works do not envisage their workplace as being potentially dangerous compared, say with a construction site. This attitude can result in accidents, and staff should be trained in the following basic aspects of personal conduct at the induction training stage.

- 4.18.7.1 Everyone should be dressed sensibly for work. Items that may catch in office machinery and unsuitable footwear should not be used.
- 4.18.7.2 Everyone should look where he/she is going. Also, reading while walking or carrying items at such a level that might block seeing should be avoided.
- 4.18.7.3 Every employee should not run or turn corners quickly when he/she cannot see what is round the corner.

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4.18.7.4 The access equipment provided should be used when storing items normally out of reach, and not revolving swivel chairs or existing shelves.

4.18.7.5 Doors should be opened carefully as there may be someone standing on the other side.

4.19 Ergonomics

Ergonomics is the science related to designing and arranging of equipment's to create safe working atmosphere. In other words, it is the science of fitting the workplace to the worker instead of forcing the worker to fit to the work place. By using the principles of ergonomics one can adjust his/her workstation (in the office or at site) to fit his/her body and avoid the stress that causes so much pain and discomfort. The following guidelines should be considered.

4.19.1 General Work Stations

4.19.1.1 The work area and tasks to be performed should be evaluated, prior to beginning work, to identify ergonomic hazards and strategize how to complete tasks safely.

4.19.1.2 The work area should be designed to permit the worker to set and read with ease, controls specific instruments. This prevents or reduces the fatigue that is so frequently the cause of faulty perception and accident.

4.19.1.3 Abnormal working postures increase the potential for fatigue, accidents and long-term injury. All work processes and systems of work should be designed to permit comfortable posture, which reduces excessive strain.

4.19.1.4 In any working area there should be free movement between operating positions, safe access and outlet, and unhindered oral and visual communication. Badly planned, congested and over-populated working areas result in operator fatigue, inattention on the part of the operator and increased accident potential in that area.

4.19.1.5 Ideally, work rates should be set to suit the operator/worker, but constant reassessment and revision is needed. It's important to note that movements, which are too slow, or too fast cause fatigue.

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- 4.19.1.6 Materials, tools and equipment should be placed in a location that reduces twisting and stretching in order to reach and grasp them.
- 4.19.1.7 An attempt has to be made to minimize handling materials in crowded and tight work areas.
- 4.19.1.8 The vertical height of items handled should be below the shoulders and above the knees.
- 4.19.1.9 Prior to performing heavy manual tasks everyone should warm up by stretching and should maintain solid footing and avoid twisting while lifting.

4.19.2 Computer Work Station

The key to comfort is in maintaining the body in a relaxed, neutral position. The ideal work position is to have the arms hanging relaxed from the shoulders. If a keyboard is used, arms should be bent at right angles at the elbow, with the hands held in a straight line with forearms and elbows close to the body. The head should be in lined with the body and slightly forward. In addition, every computer user needs to take note of the following points.

- 4.19.2.1 The height of the chair's seat should be adjusted such that the thighs are horizontal while the feet are flat on the floor.
- 4.19.2.2 The seat pan depth should be adjusted such that your back is supported by the chair back rest while the back of the knee is comfortable relative to the front of the seat.
- 4.19.2.3 The back rest should be adjusted vertically so that it supports/fits the curvature of your lower back.
- 4.19.2.4 With the arms at your sides and the elbow joint approximately 90 degrees, the height/position of the chair armrests should be adjusted to support the forearms.
- 4.19.2.5 The height of the keyboard should be adjusted such that the fingers rest on the keyboard home row when the arm is to the side, elbow at 90 degrees, and the wrist straight.
- 4.19.2.6 The mouse, trackball, or special keypads should be placed next to the keyboard tray. The wrist should be kept in a neutral position with the arm and hand close to the body.
- 4.19.2.7 The height of the monitor should be adjusted such that the top of the screen is at eye level. If bifocals/trifocals are used, the monitor should be placed at a height that allows easy viewing without tipping the head back.

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4.19.2.8 Reference documents should be placed on a document holder close to the screen and at the same distance from the eye.

4.19.2.9 A footrest may be necessary if the operator cannot rest his/her feet comfortably on the floor.

4.20 Stress at Work

Generally, a stressful circumstance is the one, which an individual is unable to cope successfully, or believes she/he cannot cope successfully, and which results in unwanted physical, mental; or emotional responses. Stress implies some form of demand on the individual, it may create physiological imbalance and can certainly affect individual performance. It is particularly concerned with how people cope with changes in their life at work, at home and in other circumstances.

As all stresses are not bad, everyone needs a certain amount of stress (positive stress) in order to cope with life situations on an on-going basis.

4.20.1 Side Effects of Stress

Work-related stress is a subject that no organization can afford to ignore for its side effects. These effects may include:

- 4.20.1.1 **Absenteeism and reduced availability for work:** -this leads to increasing costs of medical expenses; sick pay and absence cover expenses.
- 4.20.1.2 **Poor performance and reduced productivity:** -this may be below the threshold for sickness absence or reduced availability; one of the consequences of stress is reduced productivity at work.
- 4.20.1.3 **Higher accident and near-miss rates:** - Employees who are stressed or destructed are less able, or less likely, to take proper care of their own safety and that of others. This leads to higher accidents and incidents, and increasing cost of accidents.
- 4.20.1.4 **More complaints and grievances:** - Under stress, relationships become strained and small irritations and problems loom larger. The additional time and resource costs of dealing with consequences of stress, whether informally or formally are real but not quantifiable.

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4.20.1.5 ***The risk of unfair dismissal, direct or indirect:*** - An employee suffering stress to the point of distress may appear incompetent, or may feel obliged to resign in order to avoid the stress.

4.20.1.6 Given these reasons, being a good caring employer, the Company should take work place stress seriously and take sensible steps to manage it so as to protect its employees and the organization itself. It is important to note that this does not involve complex or costly arrangements; it simply calls for good management.

4.20.2 Recognizing Stress Hazards at Work

In order to put in place an integrated strategy for managing the company's potential liability for employee stress, all supervisors need to be able to identify hazard and its sources among employees.

Possible sources and causes of stress may include the following:

4.20.2.1 Stress from job

- The *job content itself*
- poor or even nonexistent task designs,
- High uncertainties about the outcomes of the job
- Unclear links between the requirements of the job and the main purpose of the organization
- Repetitive or boring tasks
- Lack of enough knowledge & skill to do the job

4.20.2.2 *The work load*

- Too much or too little responsibility
- Repetitive and monotonous manual job

4.20.2.3 *The schedule of work*

- time pressure or tight time table
- Unpredictable, long or unsocial working hours

4.20.3 Stress from the work environment

- 4.20.3.1 *Unsuitable physical work environment*
- 4.20.3.2 *Bad layout or poor workplace design*
- 4.20.3.3 *Bad housekeeping and sanitation*
- 4.20.3.4 *Poor ventilation*
- 4.20.3.5 *Excessive/weak illumination*
- 4.20.3.6 *Very high or very low temperature*
- 4.20.3.7 *Hazardous substances - dust and chemicals*

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- 4.20.3.8 Accident to others
- 4.20.3.9 Hazardous working environment in general

4.20.4 *Stress from the organization and its culture*

4.20.4.1 *The organizational structure*

- a. Unclear chain of command and accountability
- b. Ambiguity or conflict in roles
- c. Too wide or too narrow span of controls

4.20.4.2 *The organizational culture and value*

- a. The supportiveness of the culture
- b. Management's transparency, fairness and openness
- c. The way of information communication

4.20.4.3 *The amount of personal discretion allowed in the job*

- a. The extent of control of the job by the jobholder
- b. The need to decision making that the work involves and the involvement of the jobholder in decision making

4.20.4.4 *The potential for training, career development and progression*

The extent of uncertainty about job security, career progression and opportunity for advancement

4.20.4.5 *Poor interpersonal relations*

- a. The nature of the job that has to be done alone or in isolation
- b. Existence of serious interpersonal conflict with supervisors, peers or subordinate
- c. Disciplinary procedures used rigidly by inconsiderate managers
- d. Working with managers that pass on criticisms and keep the praise for them selves

4.20.4.6 *Stress caused by discrimination and harassment*

- a. Whether on the grounds of ethnicity, gender, marital status or disability, and whether it takes the form of direct or indirect discrimination
- b. Abuse of authority
- c. *Harassment of female workers by the residents of the construction site and irresponsible supervisors.*

4.20.4.7 *Stress outside the work place*

- a. Stress caused by the interaction between work and family commitment
- b. The employee's *psychological make-up*

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4.20.4.8 Stress Hazards Identification, Prevention and Control

Information from absence records, sick leaves and reported accidents should be collected and evaluated. Based on this information we need the steps below.

4.20.4.9 Treating the organization

The organization has powerful effect on the behavior of individual and groups of employees. However, the organization and its style and culture are often taken as 'given' or beyond the scope of questioning. The individual may be expected to 'take it or leave it' to adapt to it or find somewhere else to go. The organization and its culture and style have, however, real effect on health and safety.

Therefore, the organization and its management culture and style should be considered as an integral part of any co-coordinating strategy to address and manage the risk from stress at work.

4.20.4.10 Treating the job

The content of the job and its position in organization can greatly affect individual performance. Any mismatch between the requirements of the job and the capabilities and expectations of the jobholder will increase the potential for error, carelessness or lack of diligence and it will also increase the potential for stress. Therefore, job redesigning is one of the actions to be taken to control stress at work, if the assessment required it.

4.20.4.11 Treating the individual

It is difficult to predict the individual's vulnerability to stressors in the organization and in the job, and do much to modify it, but it is possible to put in place procedures and systems which will help to reduce it. These include:

- Developing job description and specifications
- Consciously trying to match the person to the job
- Providing training and supervision which enable the person to work confidently and without detailed close supervision
- Regular feedback and appraisal which should wherever possible be positive
- Creating a culture which makes it possible for employees seek help need

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f. Offering confidential counseling if the need justifies it.

5.0 Work place Accidents

An accident is an unexpected occurrence that interrupts work and usually causes injury to persons and damage to property. If this accident occurs in relation with work it is called work place accident.

5.1 Causes of Accident

5.1.1 **Unsafe Act:** Any act that deviates from a generally recognized safe of doing increases the likelihood of an accident. The basic types of unsafe acts are the following.

- a. Operating without authority
- b. Failure to make secure
- c. Operating at unsafe speed
- d. Failure to warn or signal
- e. Nullifying safety devices
- f. Using defective equipment
- g. Using equipment unsafely
- h. Taking unsafe position
- i. Repairing or servicing moving or energized equipment
- j. Riding hazardous equipment
- k. Failure to use protection
- l. Collapse of wall, parts of building (particularly during demolishing), stock of materials or stockpiles of excavated materials
- m. Collapse or over turning of ladders, scaffolds, stairs or beams
- n. Stepping on sharp objects or hand tools

5.1.2 **Unsafe Condition:** Any environmental condition that may cause or contribute to an accident. The basic types of unsafe acts are the following.

- 5.1.2.1 Inadequate guards and safety devices
- 5.1.2.2 Inadequate warning system
- 5.1.2.3 Fire and explosion hazard
- 5.1.2.4 Unexpected movement hazard
- 5.1.2.5 Poor housekeeping
- 5.1.2.6 Congestion, close clearance
- 5.1.2.7 Hazardous atmospheric condition

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- 5.1.2.8 Hazardous placement or storage
- 5.1.2.9 Unsafe equipment
- 5.1.2.10 Inadequate illumination
- 5.1.2.11 Poor ventilation
- 5.1.2.12 Hazardous personal clothing

5.1.3 Personal factor: Any condition or characteristic of a man that causes or influences him to act unsafely.

- 5.1.3.1 Lack of hazard awareness
- 5.1.3.2 Lack of job knowledge
- 5.1.3.3 Lack of job skill
- 5.1.3.4 Conflicting motivation-saving time and effort
- 5.1.3.5 Physical and mental incapacities
- 5.1.3.6 Working under the influence of alcohol or mind stimulating drugs

5.2 Types of Accidents

Work place accidents are classified mainly in to three major categories based on the level of injury and economic crises it results. These are,

5.2.1 Major accidents/injuries

These types of accidents are those which results in severe injuries, permanent disability and also death on the worker.

5.2.2 Moderate accidents/injuries

These accidents don't results permanent disability on the worker but can Cause temporally disability or injury and absent the worker days up to months from work.

5.2.3 Minor accidents /Injuries

These types of accidents results in minor injuries and the causality returns to work by receiving minor or first aid treatment and also one or two days' rest.

Common accidents that occur in the construction industry are as follows:

- a. Struck by falling objects (stone, wood, HCB, brick, hand tools, etc.)

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- b. Penetration injury by different sharp objects like nail, roofing material etc.
- c. Blunt injury by hand tool materials like hammer, chisel etc.
- d. Cutting by different materials like hand tools (saw), moving machinery parts etc.
- e. Electrical burn
- f. Falling in to open excavated poles
- g. Contact by hazardous substances or chemicals
- h. Struck against scaffolding posts and wall
- i. Contact with bare or poorly insulated electric power source and hot objects
- j. Caught in confined space or passage way, narrow manholes and machinery
- k. Caught with moving parts of a machinery
- l. Fall to different level - from scaffolding, ladders, steps, roof, moving or idle vehicle
- m. Fall on same level - tripped (stumbled) by objects such as, hand tools, raw materials, and waste materials
- n. Exposure to excessive heat, light, cold
- o. Overexertion /strain working on unsuitable (improper) position

5.3 Results (Effects) of Accidents

5.3.1 for the injured worker

- a .Permanent/ temporary disability
- b .Injury pain & Psychological impact
 - c. Loss of earning
 - d. Disrupted family
- e. Disrupted personal& social life
- f. Death etc...

5.3.2 for the organization

- 1. Injury costs (Medical costs & Sick pay)
- 2. Loss of productivity
- 3. Property damage costs

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4. Lowered employee moral
5. Lose of reputation
6. Poor customer relations
7. Lost supervisor time etc...

In order to avoid/minimize the occurrence and effects of accidents, all the concerned staffs shall follow the safety and health policies and procedures.

5.4 Reporting and Investigation of Accidents and Occupational Diseases

- 5.4.1** All employees will be held accountable for failing to report an on-the-job injury or occupational disease immediately. Immediately means at or near the time of the injury and on the same day of the injury. Employees must report the injury, in detail, to their immediate supervisor, i.e. Foreman, Site Engineer, Project administration, Project Manager or to the assigned Safety & Health officer etc. The employee must let their supervisor know:
- a. How they hurt themselves,
 - b. What they were doing at the time,
 - c. Who they were working with at the time,
 - d. When and where it happened, and,
 - e. Other pertinent information that will aid in the supervisor's investigation of the accident.
- 5.4.2** Upon receiving an accident or injury report, the responsible person shall investigate the cause and effect of the accident and report to the respective supervisor.
- 5.4.2.1** Anyone failing to report an on-the-job injury according to the above instructions is in violation of the company's Safety and Health Policy and the relevant articles of the Labor Law. Hence, a violent employee may be subject to termination. Every employee of the company, subcontractors and suppliers are expected to comply with the organization policy and the provisions of the Labor Law while working inside the company's project premises.
- 5.4.2.2** Incidents (dangerous occurrences), accidents or situations that do not necessarily result in injury to people, should also be reported immediately.

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5.5 Notification of Injuries

5.5.1 **In case of Serious Injury or Death:** After the injured has been taken to the hospital, the responsible personnel should be notified immediately. The responsible personnel has to fill out the accident report form and send it to the Head Office by the fastest possible means of communication (i.e. fax or telephone). She/he has to get statements from witnesses and make certain that the statements are signed by the witnesses, and dated and time is noted. If possible, it is advisable to take photographs of the area and anything relevant. In case of fatal injury or death it should be reported to the nearest police station.

5.5.2 **In case of Claims or Legal dispute:** The Manager of Administration&HRD should be notified in writing immediately, and He/She shall take necessary measures (consulting the GM & the legal advisor) and reply the status as fast as possible but not more than three days. (**Note:** All claims shall be handled or resolved on the basis of labor proclamation No. 377/2003.)

5.6 Prevention of Occupational accidents & Diseases

Generally we focus on three major work place accidents prevention methods. These are engineering methods, administrative controls and using PPE.

- **Engineering methods**-these methods are techniques of controlling occupational injuries and diseases at the source. There are three main categories in this method which includes elimination, substitution and separation.
 - ✓ Elimination- this method tries to avoid injuries by eliminating the hazardous working condition totally from the work place and substituting it by safe working method.
 - ✓ Substitution- substituting the hazard with less hazardous one.
 - ✓ Separation- separating the hazardous working environment from the Workers.
- **Administrative methods**- these methods are used to decrease work place injuries by using administrative strategies such as rotating workers, reducing working time (exposure time), giving awareness training ,regular medical checkup etc.
- **Using PPE**-this is a method of decreasing occupational injuries & diseases by giving different types of personal protective equipment's for workers and educating how to use it.

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Some examples of personal protective equipment's used in the construction set up-

Hand protective- Gloves

Body protective- Different protective cloths

Head protective- Safety helmet

Mouth protective- Masks

Foot protective- Safety shoe

Eye protective- Goggle etc...

6.0 Work related Welfare Facilities

6.1 First Aid and Medical Facility for Occupational Injuries

- 6.1.1.1 First aid boxes should be clearly marked and conveniently located. They should contain only first aid requisites of a prescribed standard and should be in the charge of a qualified or trained person.
- 6.1.1.2 Each material used from the first-aid box should be replaced as soon as possible.
- 6.1.1.3 Apart from first-aid boxes, it is also important to have a stretcher and a suitable rest area for injured workers.
- 6.1.1.4 When possible, hands should be washed before treating wounds, burns or eye injuries. Take care in any case not to contaminate the surfaces of dressing.
- 6.1.1.5 Sufficient percussion should be taken with regard to prevent the transmission of HIV/AIDS when treating injuries.
- 6.1.1.6 Detailed record of any case should be kept in the first-aid logbook.
- 6.1.1.7 For injuries requiring further medical treatment in clinics or hospitals a record of medical history should be kept and filed by the responsible person.

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- 6.1.1.8 When the project site is located in rural set up and if it is far from the health institutions, the company shall establish project clinic that serves for the project employees and assigned trained health professional.

6.2 Drinking-Water

Safe and adequate drinking water should be provided and maintained at convenient points, clearly marked as "Safe drinking-water". Where possible it should be provided on tap; otherwise it should be kept in suitable vessels, renewed at least daily and all practical steps taken to preserve the water from contamination.

6.3 Recreational Facilities

Recreational facilities offer workers the opportunity to spend their leisure time in activities likely to increase physical and mental well-being. They also help to improve social relations within the organization. Therefore facilities for outdoor sports, such as volleyball, table tennis and for indoor games, such as chase, dart, etc. should be provided as much as possible. Furthermore, the company should organize and subsidize picnics, social events or holidays giving special attention to the needs of female workers.

6.4 HIV/AIDS prevention campaigns

Apart from the safety & health activities, the company tries to work on the prevention of this global pandemic by awareing the employees about the nature of the epidemic, distributing condoms at site and supporting the victims financially and psychologically.

7.0Recording and Documentations

7.1 Major Accident or Incident Investigation and Report Form

This form is used to report accidents or incidents of a major nature to the Head Office. Victims of major accidents or incidents are those who can't resume their work after being given first-aid at site. Although they might not cause injury to

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a human being, vehicle, construction equipment accidents caused damage should be considered and reported as major accidents. The responsibility to ensure the timely reporting of major accidents or incidents at project sites is that of the HPAS/SHO.

7.2 Work place accidents Registration Form

This form is used to bring the occurrence of accidents in the workplace. The report should registered and reported weekly/monthly by HPAS and/or SHO to the HO.

7.3 Medical Treatment Follow up form

This register is kept for each causality that has been undergoing through a medical treatment. The SHO at HO and HPAS and/or SHO at sites are responsible to maintain medical treatment follow up sheet.

7.4 Monthly Safety & Health activities Reporting Format

The monthly safety & health activities reporting format is a summary of main & detailed safety & health activities done by the responsible personnel and reported monthly and/or weekly as necessary by the HPAS and/or SHO at PO and to be compiled by the HRDAD/SHO at HO.

7.5 Work place accidents Reports summary form

This form is used to report the summary of work place accidents and related information that was happened during the reporting period .this form is filled and submitted by HPAS and/or SHO at PO and to be compiled by the HRDAD/SHO at HO.

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7.6 Patient Referral Sheet

This form is used to refer cases which are beyond the project clinic to the higher health institution by the HPAS and /or project nurse, Health Officer or SHO for better diagnosis and treatment.

7.7 work place accidents report summary form(HO)

This form is used to report the summary of work place accidents that was happened in all projects during the reporting period.it is filled and compiled by the HRDAD/SHO at HO.

7.8 Safety & Health activities report form(HO)

This is used to report the safety & health activities done during the specified period at the company level.it is filled &reported by the HRDAD/SHO at HO.

8.0 Safety Monitoring System (SMS)

Safety monitoring is concerned with the measurement and evaluation of safety performance. The organization should exert every effort required to ensure that the safety and health system is implemented and timely updated. Safety monitoring should be carried out in the company through the following methods.

8.1 Safety Surveys

This is a detailed examination of a number of critical areas of operation or, perhaps, an in-depth study of all health and safety related activities in a workplace. Safety surveys should be conducted by the HRDAD, OSHC and SHO at HO and HPAS, SHO and SOSHC at PO.

8.2 Safety Tours

These are unscheduled examination of working area, frequently undertaken and a group exercise (e.g. by Forman, safety representative and safety committee members), to assess general compliance with safety requirements (e.g. housekeeping, access at site and use of machinery safety devices).

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8.3 Safety Audit and Inspection

A safety audit fundamentally subjects each area of an organization's activities to a systematic critical examination with the object of minimizing injury and loss. It generally takes the form of a series of questions directed to examining factors such as the operation of safe system of work, compliance with the statement of Health and Safety Policy, levels of legal compliance and the operation of hazard reporting systems.

Safety audit should be conducted by the HRDAD, OSHC and SHO at HO and by the PM, SOSHC, HPAS and SHO at PO, using Safety Audit Checklist that should be developed by the concerned staff.

8.4 Contact Schemes

It is a system for imparting information and/or instructions on safety procedures, whereby the HRDAD, OSHC and SHO at HO and HPAS, SHO and SOSHC at PO undertake planned contact with individual members of the workforce to discuss or reinforce particular safety topics.

8.5 Hazard and Operability Studies

The principal aim of such a study is to assess the hazard potential arising from the incorrect operation of equipment and consequential effects on the facility. Such an operation enables remedial action to be taken at a very early stage and should be undertaken by the HRDAD, OSHC and SHO at HO and HPAS, SHO and SOSHC at PO jointly with Engineering Department responsible personnel.

8.6 Management Committee Monitoring

The monitoring by the Management Committee particularly focused on the implementation of the safety and health policy, follow-up of recommendations given by the HRDAD/PM, OSHC/SOSHC and/or concerned supervisors with regard to safety and health etc. that should help maintain safe and hazard free working environment allover the Company.

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